INCH-POUND

MS39212K
31 January 2018
SUPERSEDING
MS39212J
30 August 2012

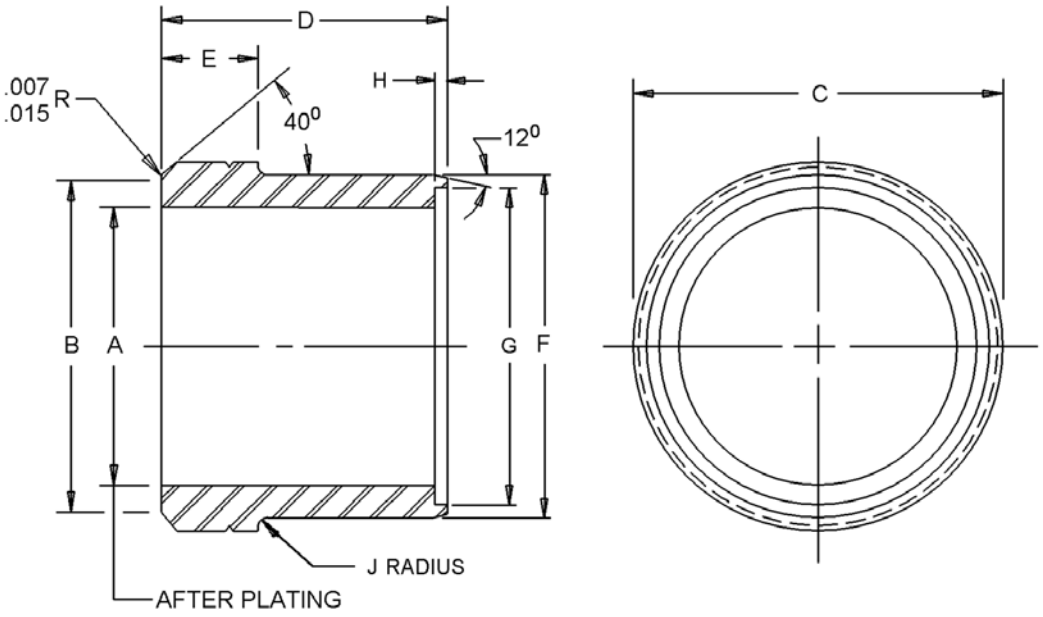
DETAIL SPECIFICATION SHEET

SLEEVES, COMPRESSION, TUBE - HOSE FITTING- SAFETY SLEEVE TYPE

Reactivated after 31 January 2018 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.



Inches	mm
.007	0.18
.015	0.38

FIGURE 1. Sleeve, compression, tube.

AMSC N/A

FSC 4730

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Dash number	Tube OD Inches (mm)	A dia. Inches (mm)	B dia. Inches (mm)	C Inches (mm)	D Inches (mm)
-1	.125 (3.18)	.129 (3.28) .131 (3.33)	.160 (4.06) .150 (3.81)	.213 (5.41) .203 (5.16)	.281 (7.14) .275 (6.99)
-2	.188 (4.78)	.190 (4.83) .193 (4.90)	.250 (6.35) .230 (5.84)	.322 (8.18) .312 (7.92)	.281 (7.14) .275 (6.99)
-3	.250 (6.35)	.255 (6.48) .258 (6.55)	.316 (8.03) .296 (7.52)	.380 (9.65) .370 (9.40)	.339 (8.61) .333 (8.46)
-4	.313 (7.95)	.318 (8.08) .321 (8.15)	.379 (9.63) .359 (9.12)	.448 (11.38) .438 (11.13)	.339 (8.61) .333 (8.46)
-5	.375 (9.53)	.380 (9.65) .383 (9.73)	.441 (11.20) .421 (10.69)	.505 (12.83) .495 (12.57)	.378 (9.60) .372 (9.45)
-6	.500 (12.70)	.506 (12.85) .509 (12.93)	.567 (14.40) .547 (13.89)	.635 (16.13) .625 (15.88)	.378 (9.60) .372 (9.45)
-7	.625 (15.88)	.633 (16.08) .636 (16.15)	.694 (17.63) .674 (17.12)	.760 (19.30) .750 (19.05)	.418 (10.62) .412 (10.46)
-8	.750 (19.05)	.758 (19.25) .761 (19.33)	.819 (20.80) .799 (20.29)	.885 (22.48) .875 (22.23)	.418 (10.62) .412 (10.46)
-9	.875 (22.23)	.883 (22.43) .886 (22.50)	.944 (23.98) .924 (23.47)	1.010 (25.65) 1.000 (25.40)	.418 (10.62) .412 (10.46)
-10	1.000 (25.40)	1.008 (25.60) 1.011 (25.68)	1.069 (27.15) 1.049 (26.64)	1.135 (28.83) 1.125 (28.58)	.418 (10.62) .412 (10.46)
-11	1.250 (31.75)	1.261 (32.03) 1.264 (32.11)	1.322 (33.58) 1.302 (33.07)	1.416 (35.97) 1.406 (35.71)	.418 (10.62) .412 (10.46)
-12	1.500 (38.10)	1.511 (38.38) 1.514 (38.46)	1.572 (39.93) 1.552 (39.42)	1.666 (42.32) 1.656 (42.06)	.418 (10.62) .412 (10.46)

FIGURE 1. Sleeve, compression, tube - Continued.

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Dash number	E Inches (mm)	F dia. Inches (mm)	G dia. ± .005 inch (0.13 mm) Inches (mm)	H Inches (mm)	J radius ± .005 inch (0.13 mm) Inches (mm)
-1	.084 (2.13) .078 (1.98)	.178 (4.52) .175 (4.45)	.140 (3.56)	.015 (0.38) .023 (0.58)	.010 (.25)
-2	.084 (2.13) .078 (1.98)	.256 (6.50) .252 (6.40)	.205 (5.21)	.020 (0.51) .028 (0.71)	.020 (0.51)
-3	.102 (2.59) .096 (2.44)	.338 (8.59) .334 (8.48)	.268 (6.81)	.024 (0.61) .038 (0.97)	.020 (0.51)
-4	.102 (2.59) .096 (2.44)	.400 (10.16) .396 (10.06)	.330 (8.38)	.024 (0.61) .038 (0.97)	.020 (0.51)
-5	.122 (3.10) .116 (2.95)	.461 (11.71) .457 (11.61)	.393 (9.98)	.024 (0.61) .038 (0.97)	.020 (0.51)
-6	.122 (3.10) .116 (2.95)	.589 (14.96) .585 (14.86)	.521 (13.23)	.024 (0.61) .038 (0.97)	.020 (0.51)
-7	.122 (3.10) .116 (2.95)	.715 (18.16) .711 (18.06)	.647 (16.43)	.024 (0.61) .038 (0.97)	.020 (0.51)
-8	.122 (3.10) .116 (2.95)	.840 (21.34) .836 (21.23)	.772 (19.61)	.024 (0.61) .038 (0.97)	.020 (0.51)
-9	.122 (3.10) .116 (2.95)	.965 (24.51) .961 (24.41)	.897 (22.78)	.024 (0.61) .038 (0.97)	.020 (0.51)
-10	.122 (3.10) .116 (2.95)	1.090 (27.69) 1.086 (27.58)	1.022 (25.96)	.024 (0.61) .038 (0.97)	.020 (0.51)
-11	.122 (3.10) .116 (2.95)	1.344 (34.14) 1.340 (34.04)	1.275 (32.39)	.024 (0.61) .038 (0.97)	.020 (0.51)
-12	.122 (3.10) .116 (2.95)	1.593 (40.46) 1.589 (40.36)	1.524 (38.71)	.024 (0.61) .038 (0.97)	.020 (0.51)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance for degrees is ± 2°.
4. This illustration is for identification and is not intended to restrict designs or shapes not dimensioned.

FIGURE 1. Sleeve, compression, tube - Continued.

REQUIREMENTS:

Dimensions and configuration: Fittings shall be as specified on figure 1 and table I.

Material: Carbon steel and corrosion resistant steel shall be in accordance with SAE-J514 for flareless type ferrules.

Finish: Finishes shall be as specified in table I. All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages and other openings shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

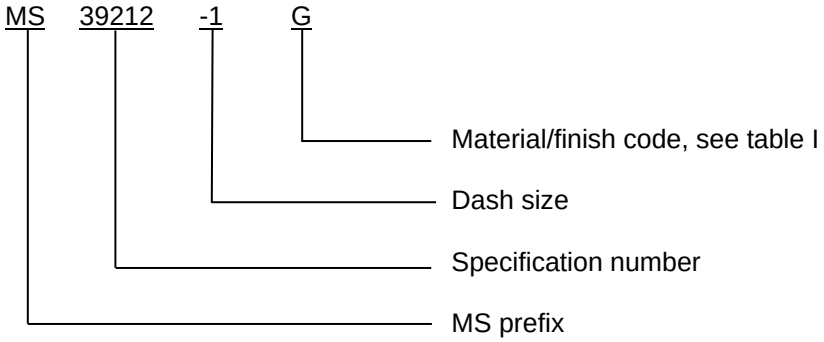
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TABLE I. Material and finish designators.

Designator	Material	Finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/
F	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 2/
G	Steel	Zinc plating with chromate conversion in accordance with ASTM B695, type II, class 8. 2/
H	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 2/
J	Steel	Zinc plating with chromate conversion in accordance with ASTM B633; type VI, Fe/Zn 8. 2/
P	Steel	Zinc plating with colorless passivate in accordance with ASTM B633; type III or V, Fe/Zn 8. 2/
S	Corrosion Resistant steel 3/	No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7. 3/
V	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 4/ 5/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA owned shipboard systems.
2/ Not for use in aircraft. Requires approval from the Program Office for all applications.
3/ Stainless steel of such hardness as to be capable of biting, fully annealed type 304 stainless steel tubing.
4/ Hexavalent chromium free. Finish shall be RoHS compliant.
5/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.

Part or Identifying Number (PIN) example:



MS39212-1G indicates a sleeve compression for a .125 (3.18) tube, steel, zinc plating with chromate conversion.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2.

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Class I and II ozone depleting substances (ODS) shall not be used in MS39212 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. This document references the following:

MIL-DTL-16232	SAE-AMS2417
ASTM B117	SAE-AMS2700
ASTM B633	SAE-AMS-QQ-P-416
ASTM B695	SAE-J514
ASTM F1136/F1136M	T9070-AL-DPC-020/077-2
SAE-AMS-C-81562	

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2018-018)

Review activities:
Army - AR
Navy - AS, MC, SA
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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